

RAJALAKSHMI ENGINEERING COLLEGE

An Autonomous Institution, Affiliated to Anna University

Rajalakshmi Nagar, Thandalam – 602 105

Web Technology And Mobile Application

Experiment 6

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By

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AIM:

To develop a servlet-based Online Railway Reservation System that demonstrates **session tracking using** HttpSession by implementing a simple login, dashboard, and logout mechanism.

ALGORITHM:

1. Read the action **parameter** from request to control flow (login, dashboard, logout).
2. If action = login:
 1. Get username and password using request.getParameter()
 2. Validate input (user and 1234)
 3. If valid:
 1. Create session using request.getSession()
 2. Store username using session.setAttribute()
 4. Else:
 1. Show “Invalid Credentials”
3. If action = dashboard:
 1. Get existing session using request.getSession(false)
 2. If session exists:
 1. Retrieve username using session.getAttribute()
 2. Display dashboard
 3. Else:
 1. Show “Session Expired”
4. If action = logout:
 1. Get session using request.getSession(false)
 2. Invalidate session using session.invalidate()
 3. Redirect to login page
5. If no action:
 1. Display login form
6. Use PrintWriter to generate **HTML + CSS output inside servlet**

CODE:

Servlet code:

```
package wtma;
```

```
import jakarta.servlet.ServletException;  
import jakarta.servlet.annotation.WebServlet;  
import jakarta.servlet.http.*;  
import java.io.IOException;  
import java.io.PrintWriter;
```

```
@WebServlet("/exp6")  
public class exp6 extends HttpServlet {  
    private static final long serialVersionUID = 1L;
```

```
    protected void doGet(HttpServletRequest request, HttpServletResponse response)  
        throws ServletException, IOException {  
        processRequest(request, response);  
    }
```

```
    protected void doPost(HttpServletRequest request, HttpServletResponse response)  
        throws ServletException, IOException {  
        processRequest(request, response);  
    }
```

```
    private void processRequest(HttpServletRequest request, HttpServletResponse  
        response)  
        throws IOException {
```

```
        response.setContentType("text/html");  
        PrintWriter out = response.getWriter();
```

```
        String action = request.getParameter("action");
```

```
        // ===== LOGIN =====  
        if ("login".equals(action)) {
```

```
            String username = request.getParameter("username");  
            String password = request.getParameter("password");
```

```
            if (username != null && password != null) {  
                username = username.trim();  
                password = password.trim();
```

```
                if ("Navdeep".equalsIgnoreCase(username) && "Nn@1234".equals(password)) {
```

```
HttpSession session = request.getSession();
session.setAttribute("username", username);
```

```
out.println(pageStart());
out.println("<div class='card'>");
out.println("<h2>Login Successful</h2>");
out.println("<a class='btn' href='exp6?action=dashboard'>Go to Dashboard</a>");
out.println("</div>");
out.println(pageEnd());
```

```
} else {
out.println(pageStart());
out.println("<div class='card'>");
out.println("<h2 style='color:red;'>Invalid Credentials</h2>");
showLoginForm(out);
out.println("</div>");
out.println(pageEnd());
}
}
```

```
// ===== DASHBOARD =====
else if ("dashboard".equals(action)) {
```

```
HttpSession session = request.getSession(false);
```

```
if (session != null && session.getAttribute("username") != null) {
```

```
String user = (String) session.getAttribute("username");
```

```
out.println(pageStart());
out.println("<div class='card'>");
out.println("<h2>Welcome, " + user + "</h2>");
out.println("<p>Select an option:</p>");
```

```
out.println("<ul style='color: black;'>");
out.println("<li> Book Ticket</li>");
out.println("<li>✕ Cancel Ticket</li>");
out.println("<li> Check Status</li>");
out.println("</ul>");
```

```
out.println("<a class='btn logout' href='exp6?action=logout'>Logout</a>");
out.println("</div>");
out.println(pageEnd());
```

```
} else {
```

```
out.println(pageStart());
out.println("<div class='card'>");
out.println("<h2>Session Expired</h2>");
showLoginForm(out);
out.println("</div>");
out.println(pageEnd());
}
```

```
// ===== LOGOUT =====
else if ("logout".equals(action)) {
```

```
HttpSession session = request.getSession(false);
```

```
if (session != null) {
    session.invalidate();
}
```

```
out.println(pageStart());
out.println("<div class='card'>");
out.println("<h2>Logged out successfully</h2>");
showLoginForm(out);
out.println("</div>");
out.println(pageEnd());
}
```

```
// ===== DEFAULT =====
else {
    out.println(pageStart());
    out.println("<div class='card'>");
    showLoginForm(out);
    out.println("</div>");
    out.println(pageEnd());
}
```

```
// ===== COMMON PAGE START =====
private String pageStart() {
    return "<html><head><title>Railway System</title>"
+ "<style>"
+ "body{margin:0;font-family:Segoe UI,Arial;background-image:url('background"
+ " orrs.avif');background-size: cover;\r\n"
+ " background-position: center;\r\n"
+ " background-repeat: no-repeat;\r\n"
+ " background-attachment: fixed;\r\n"
+ " background-color: whitesmoke;}"
```

```

+ ".header{background:#222;color:white;padding:15px;text-align:center;font-
size:22px;font-weight:bold;}"
+ ".container{display:flex;justify-content:center;align-items:center;height:90vh;}"
+ ".card{background:black;color:white;padding:30px;border-radius:12px;"
+ "box-shadow:0 8px 20px rgba(0,0,0,0.3);text-align:center;width:320px;"
+ "transition:transform 0.3s;}"
+ ".card:hover{transform:scale(1.03);}"
+ "input{width:90%;padding:10px;margin:10px 0;border-radius:6px;border:1px solid
#ccc;}"
+ "input:focus{border-color:#667eea;outline:none;}"
+ ".btn{display:inline-block;padding:10px 18px;margin-top:10px;"
+ "background:#667eea;color:white;text-decoration:none;border-radius:6px;"
+ "font-weight:bold;}"
+ ".btn:hover{background:#5a67d8;}"
+ ".logout{background:#e53e3e;}"
+ ".logout:hover{background:#c53030;}"
+ "ul{list-style:none;padding:0;margin-top:15px;}"
+ "li{background:#f4f4f4;margin:6px 0;padding:8px;border-radius:5px;}"
+ "</style></head><body>"
+ "<div class='header'> Online Railway Reservation System</div>"
+ "<div class='container'>";
}

```

```

// ===== COMMON PAGE END =====
private String pageEnd() {
return "</div></body></html>";
}

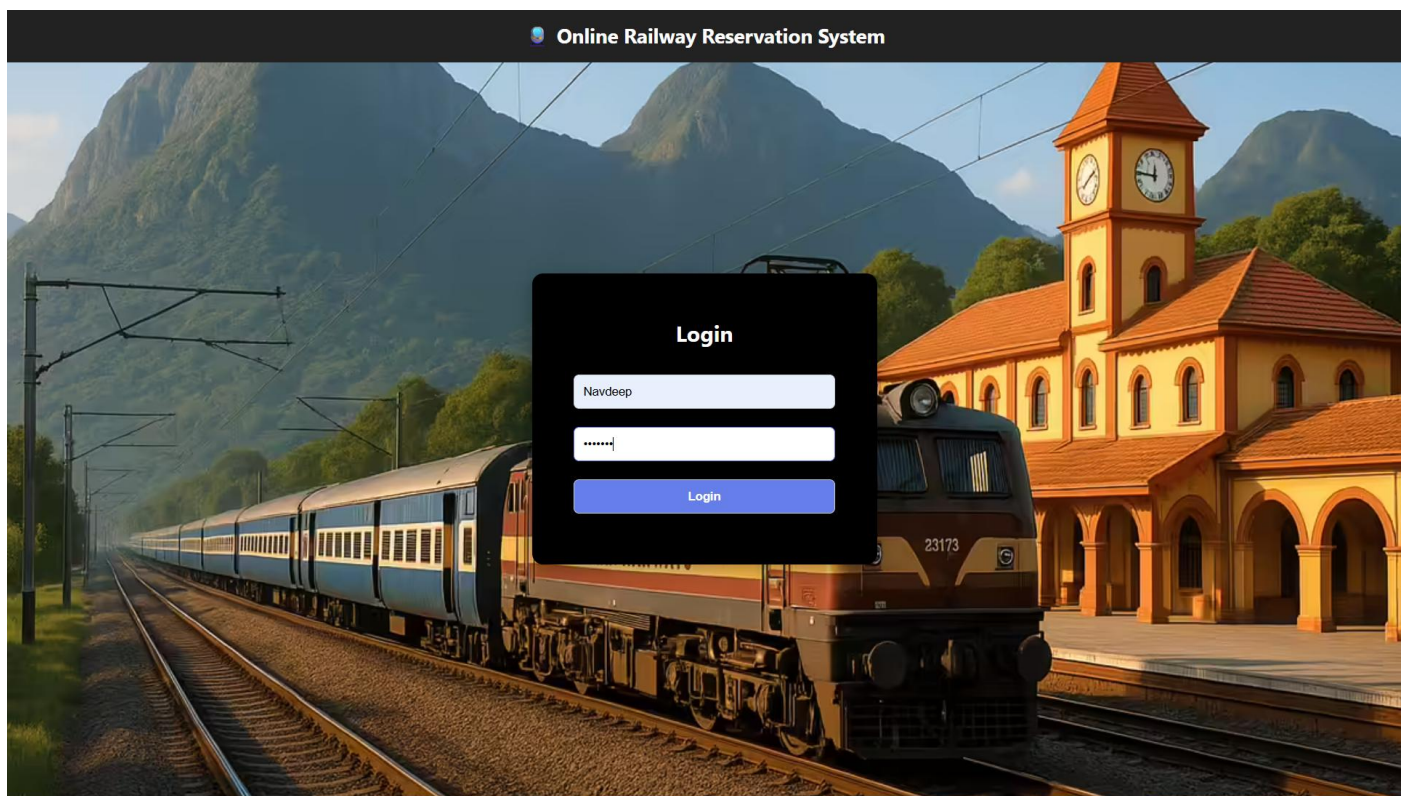
```

```

// ===== LOGIN FORM =====
private void showLoginForm(PrintWriter out) {
out.println("<h2>Login</h2>");
out.println("<form action='exp6' method='post'>");
out.println("<input type='hidden' name='action' value='login'>");
out.println("<input type='text' name='username' placeholder='Username'><br>");
out.println("<input type='password' name='password'
placeholder='Password'><br>");
out.println("<input class='btn' type='submit' value='Login'>");
out.println("</form>");
}
}

```

OUTPUT:



Login Successful

[Go to Dashboard](#)

Welcome, Navdeep

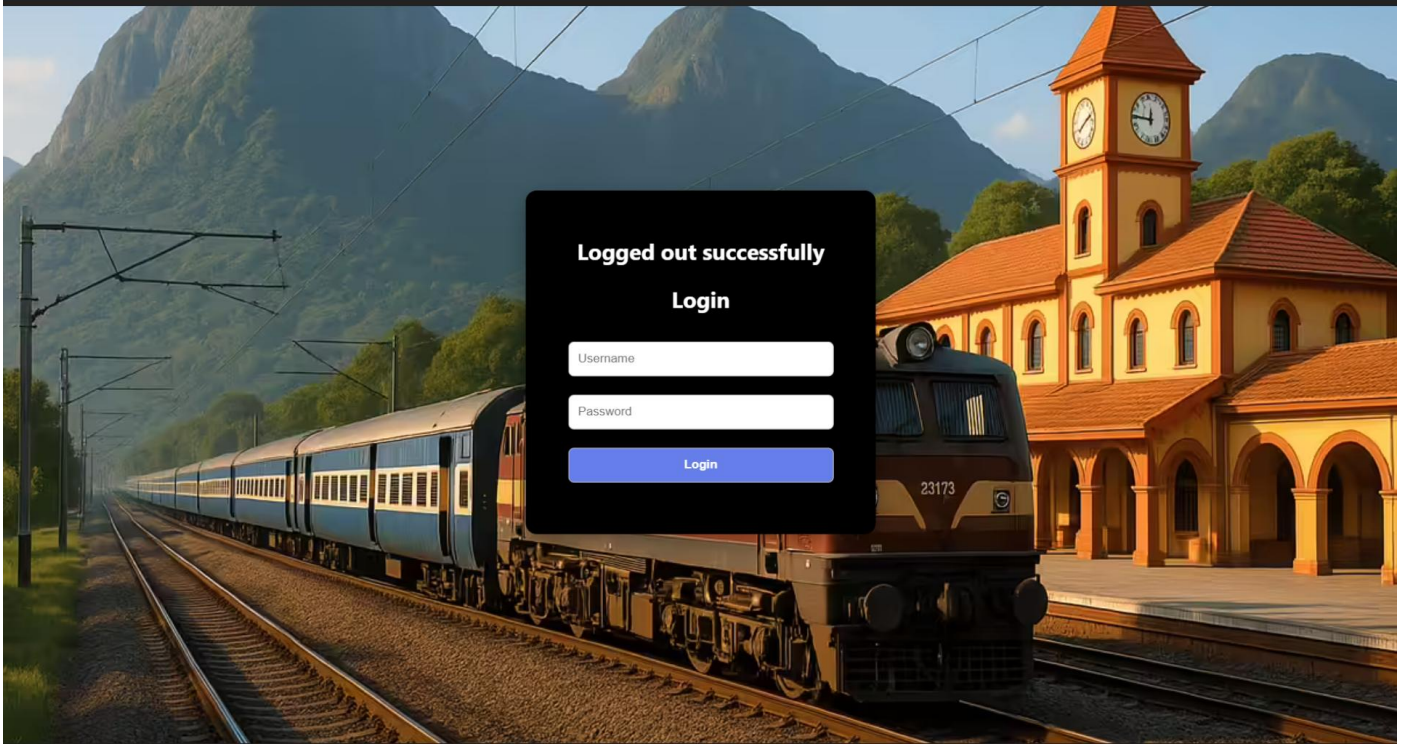
Select an option:

[Book Ticket](#)

[Cancel Ticket](#)

[Check Status](#)

[Logout](#)



RESULT:

The servlet successfully demonstrates **session tracking using** HttpSession. User logs in → session is created → user data is maintained across pages → on logout, session is destroyed.